

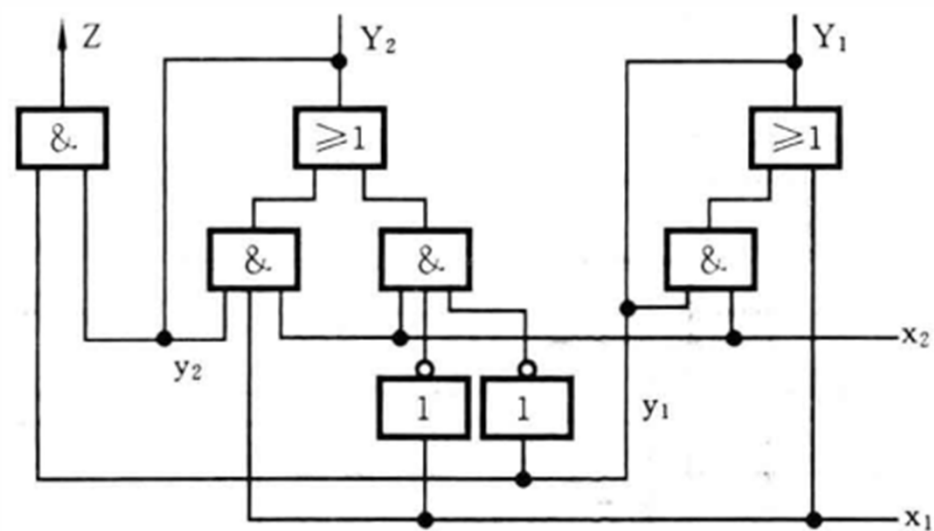
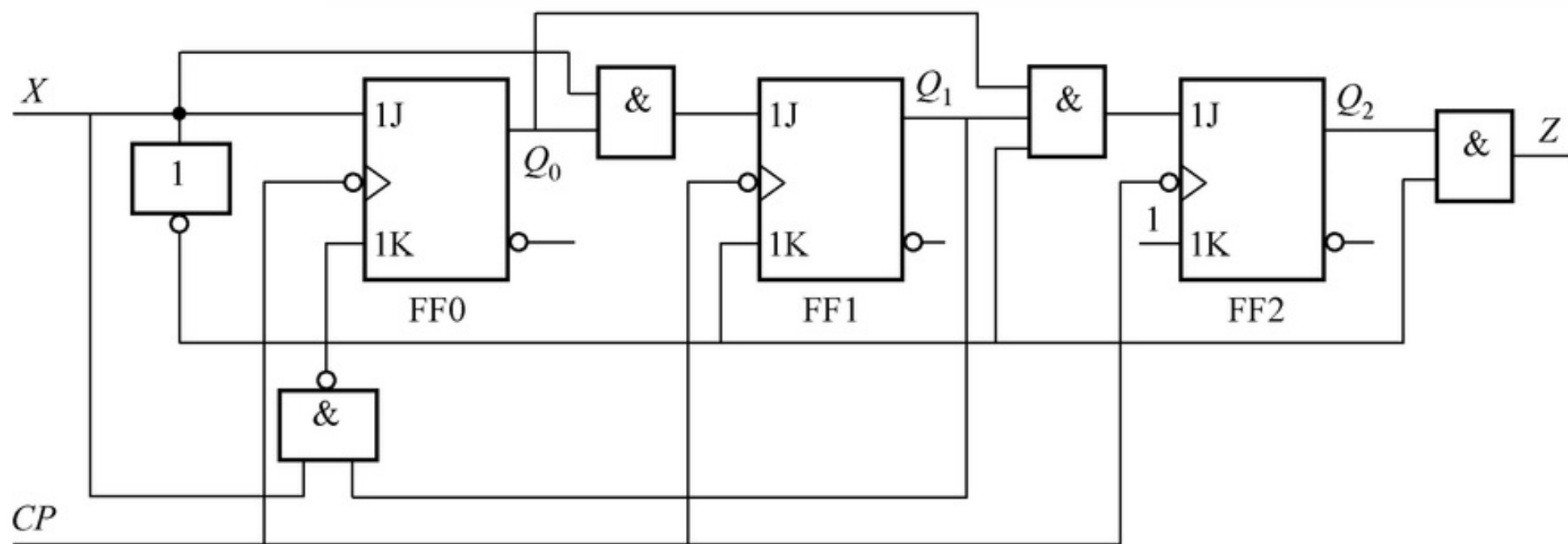


图 6.17 逻辑电路



廈門大學
XIAMEN UNIVERSITY

机器学习导论

第十四讲：强化学习基础



人工智能系

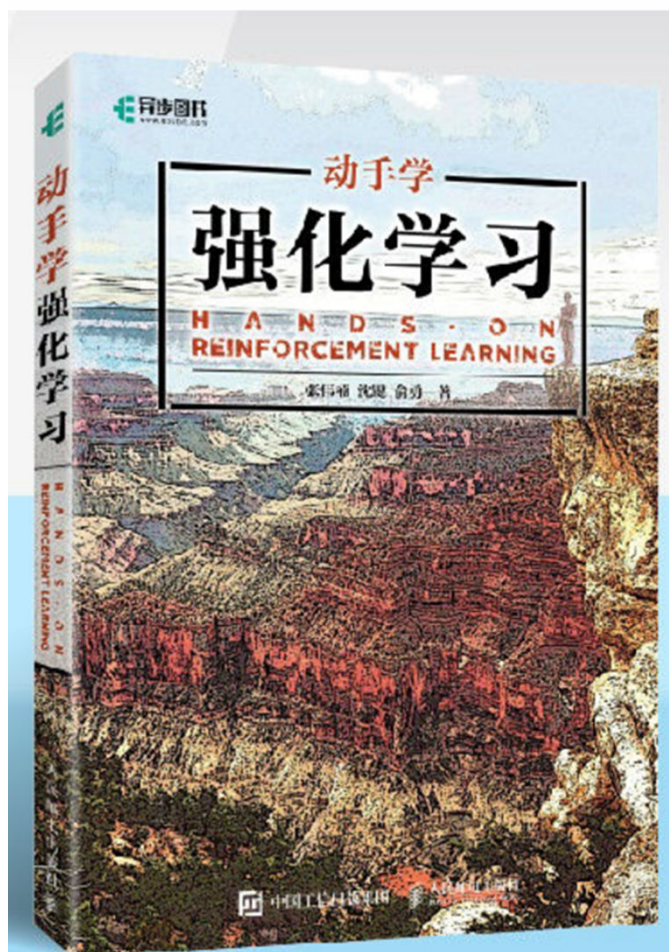
晁飞副教授

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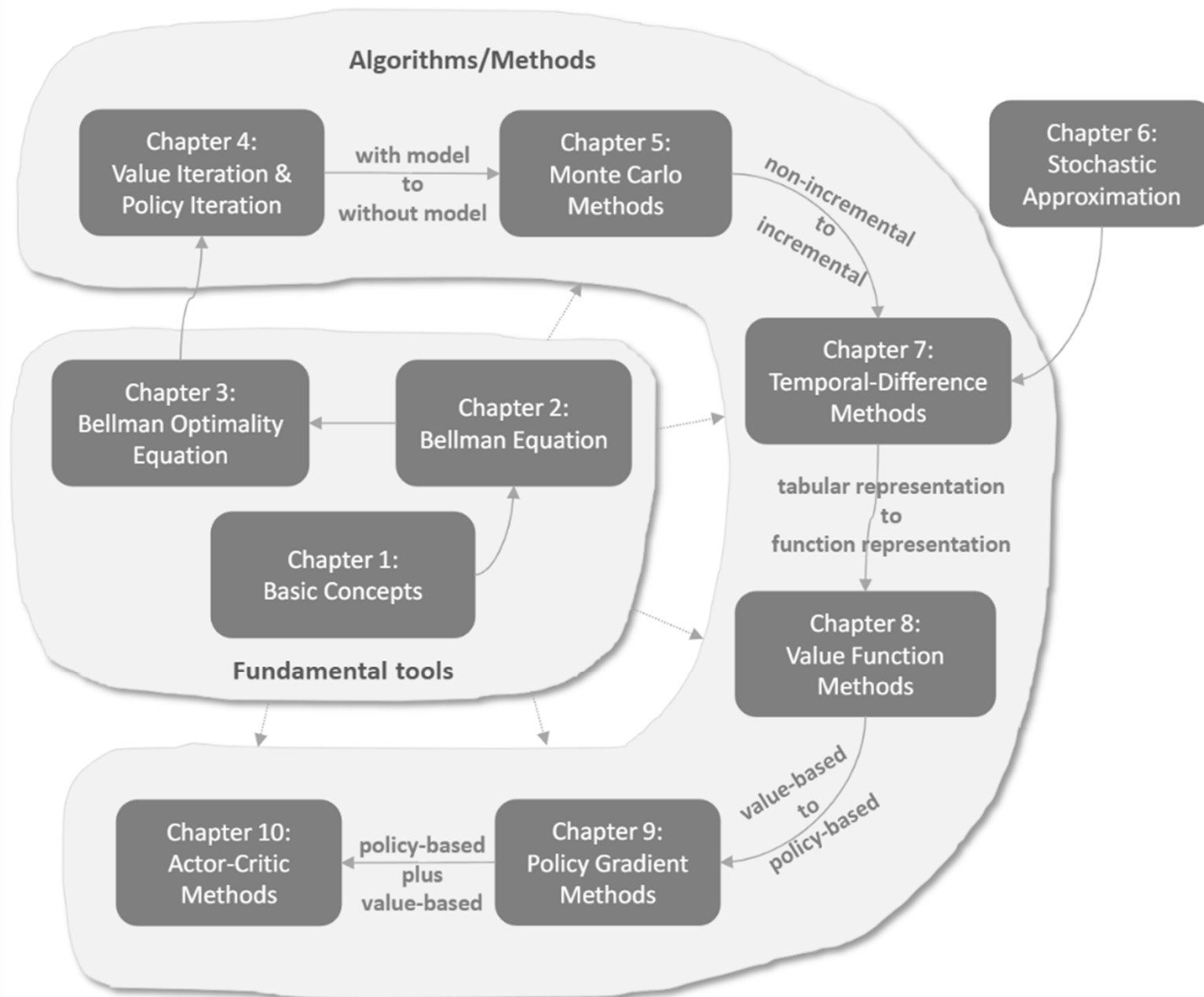
目录

- 基本概念、术语
- 基于动作价值的强化学习
- 基于策略的强化学习
- Actor-critic方法和我自己的研究内容

书籍推荐

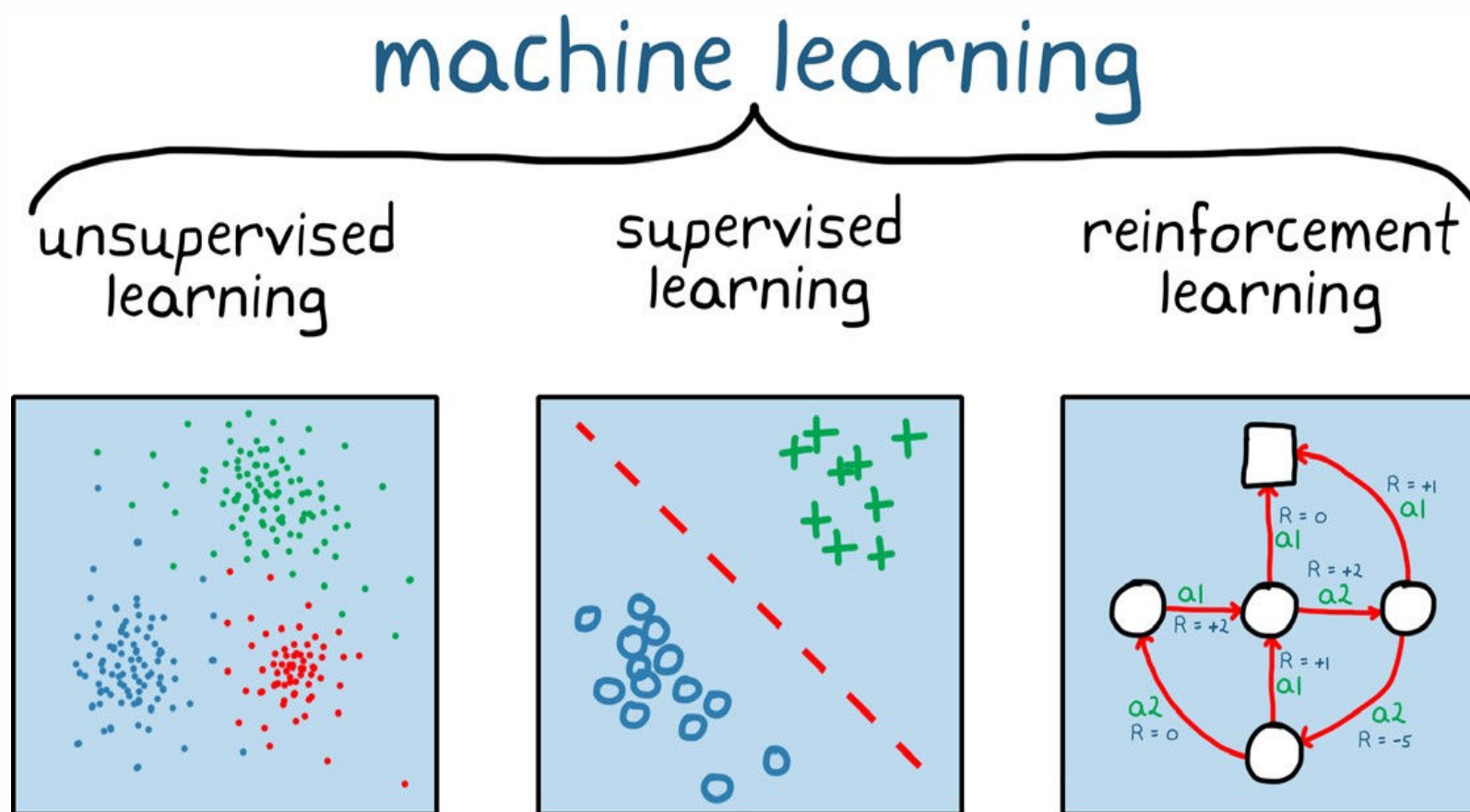


赵世钰老师的书



强化学习：机器学习的一个子集

强化学习是机器学习的三大类之一。
先了解强化学习与这两者的不同之处。



强化学习与这两者的不同之处

Supervised Learning

Data: (x, y)

x is data, y is label

Goal: Learn function to map
 $x \rightarrow y$

Apple example:



This thing is an apple.

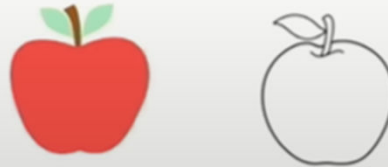
Unsupervised Learning

Data: x

x is data, no labels!

Goal: Learn underlying
structure

Apple example:



This thing is like
the other thing.

Reinforcement Learning

Data: state-action pairs

Goal: Maximize future rewards
over many time steps

Apple example:



Eat this thing because it
will keep you alive.

强化学习的专业术语

术语: Agent



AGENT

Agent: takes actions.



ENVIRONMENT

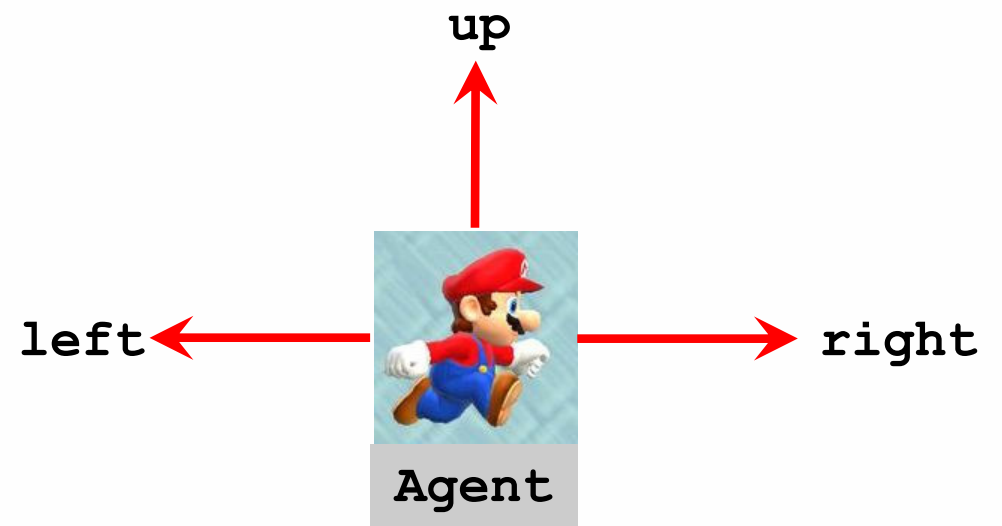
Environment: the world in which the agent exists and operates.

术语: state and action

state s (this frame)



Action $a \in \{\text{left}, \text{right}, \text{up}\}$

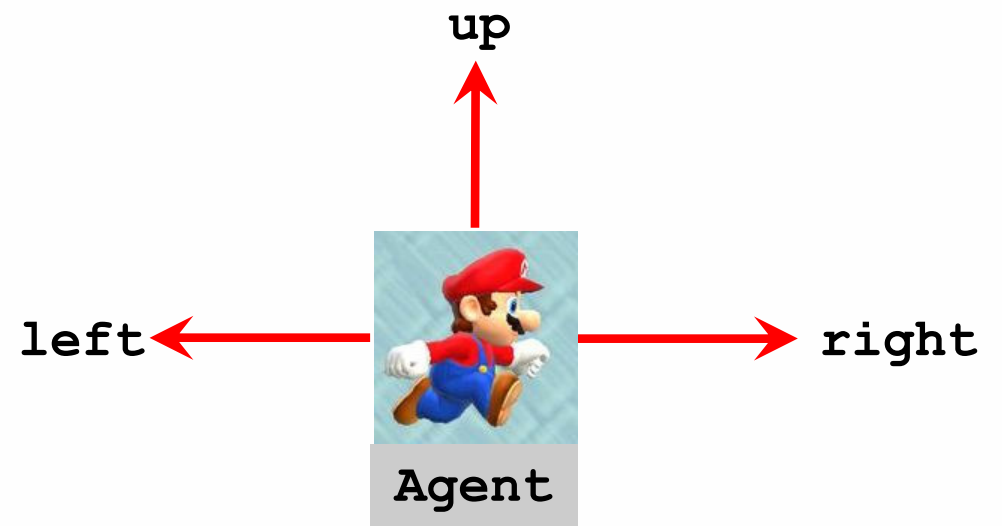


术语: state and action

state s (this frame)



Action $a \in \{\text{left}, \text{right}, \text{up}\}$

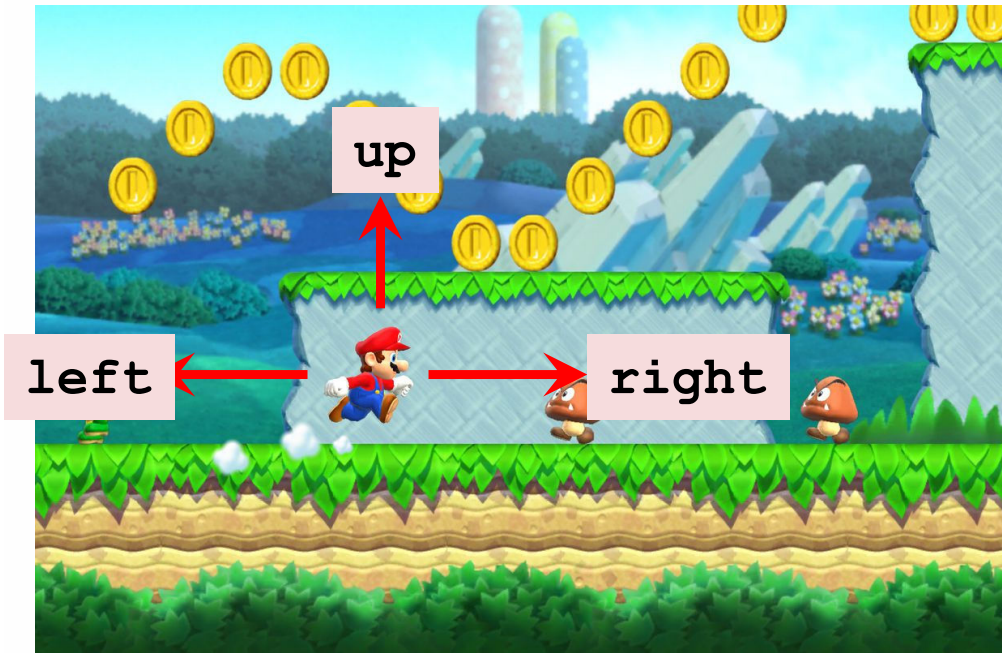


术语: policy

policy π

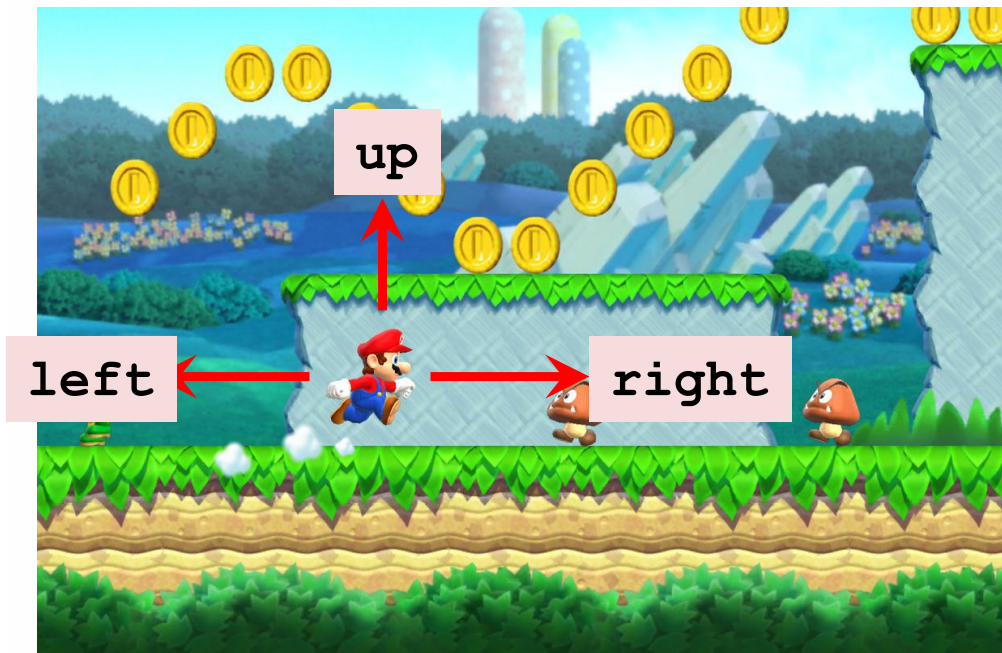
- Policy function $\pi: (s, a) \mapsto [0,1]$:

$$\pi(a | s) = \mathbb{P}(A = a | S = s).$$



术语: policy

policy π

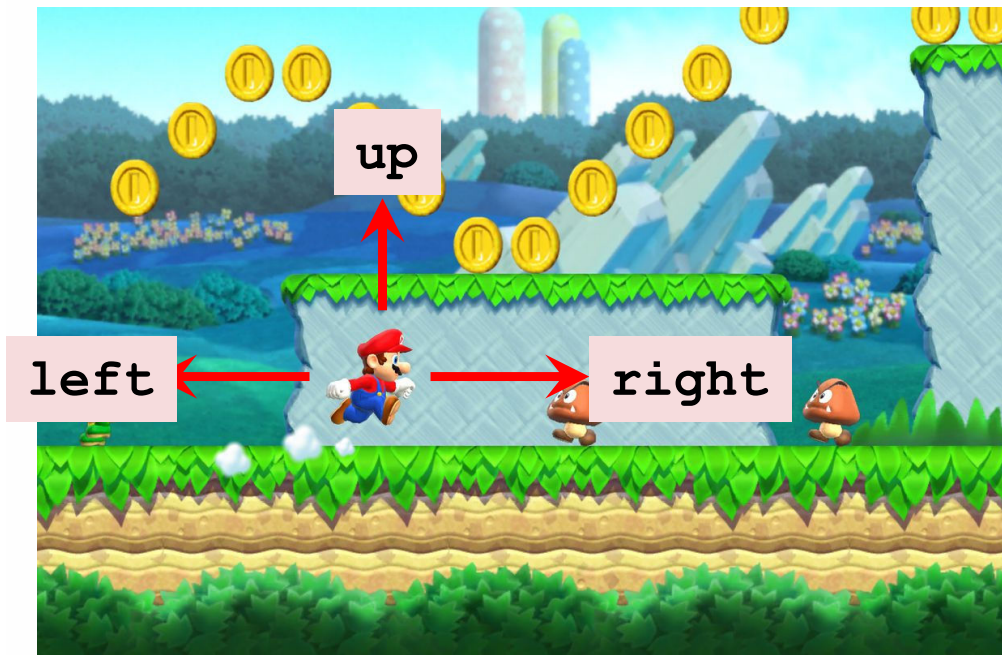


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术语: policy

policy π

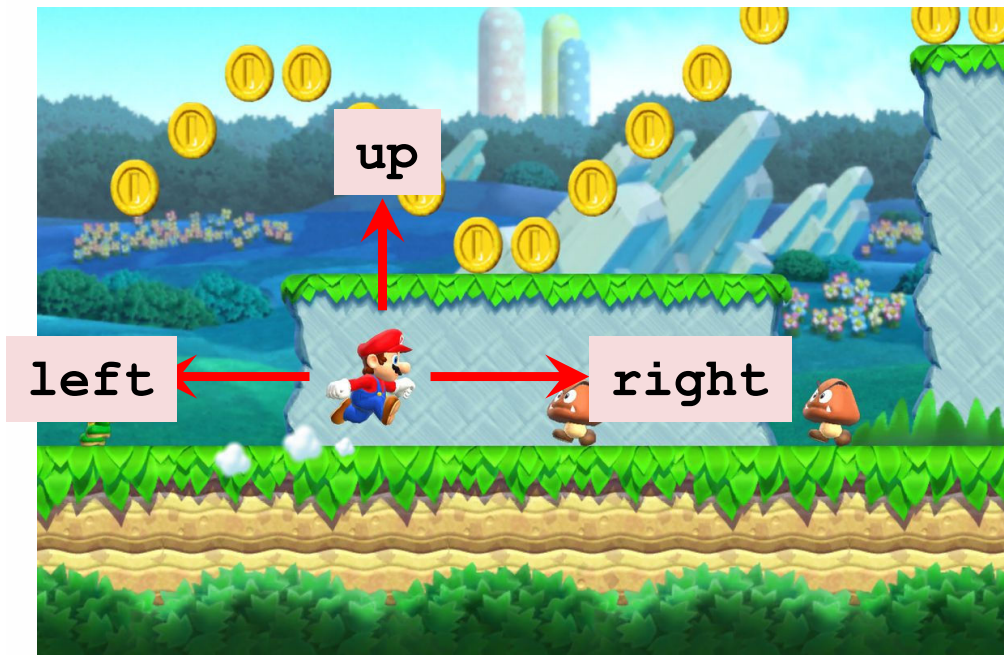


- Policy function $\pi: (s, a) \mapsto [0,1]$:

$$\pi(a | s) = \mathbb{P}(A = a | S = s).$$

术语: policy

policy π

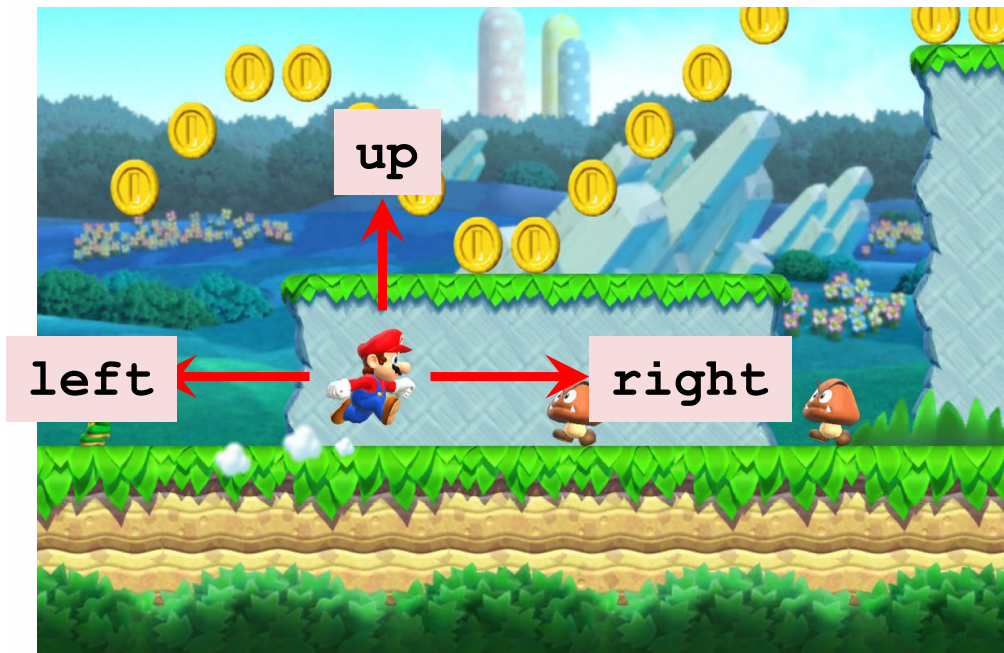


- $\pi(a | s)$ is the probability of taking action $A = a$ given state s , e.g.,

- ➔ • $\pi(\text{left} | s) = 0.2$,
- ➔ • $\pi(\text{right} | s) = 0.1$,
- ➔ • $\pi(\text{up} | s) = 0.7$.

术语: policy

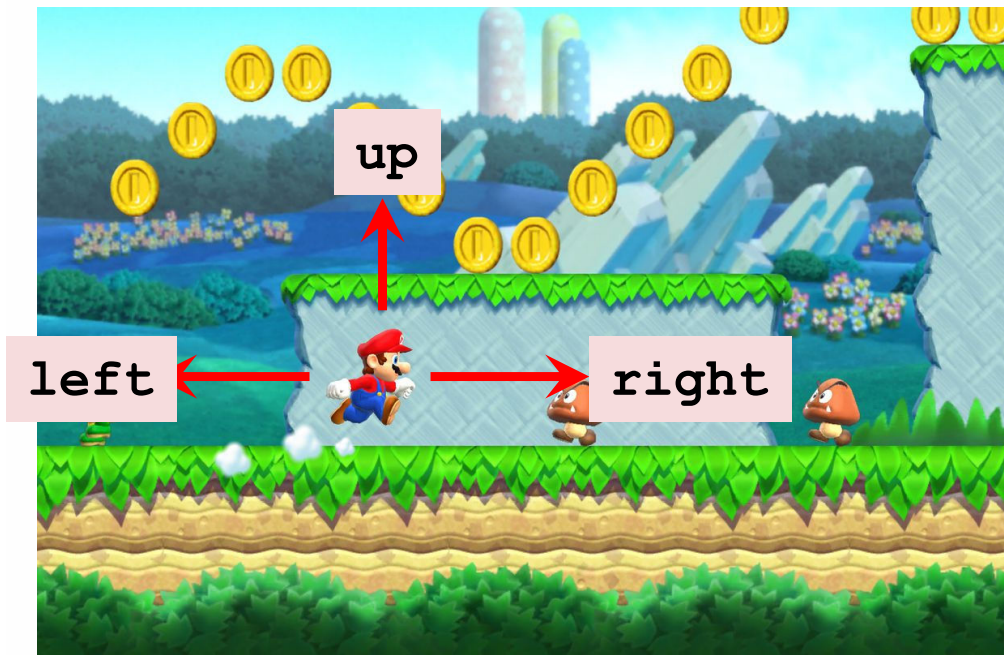
policy π



- $\pi(a | s)$ is the probability of taking action $A = a$ given state s , e.g.,
 - $\pi(\text{left} | s) = 0.2$,
 - $\pi(\text{right} | s) = 0.1$,
 - $\pi(\text{up} | s) = 0.7$.
- Upon observing state $S = s$, the agent's action A can be random.

术语: policy

Random or deterministic policy?



术语: reward

reward R

- Collect a coin: $R = +1$



术语: reward



reward R

- Collect a coin: $R = +1$
- Win the game: $R = +10000$

术语: reward



reward R

- Collect a coin: $R = +1$
- Win the game: $R = +10000$
- Touch a Goomba: $R = -10000$ (game over).

术语: reward



reward R

- Collect a coin: $R = +1$
- Win the game: $R = +10000$
- Touch a Goomba: $R = -10000$ (game over).
- Nothing happens: $R = 0$

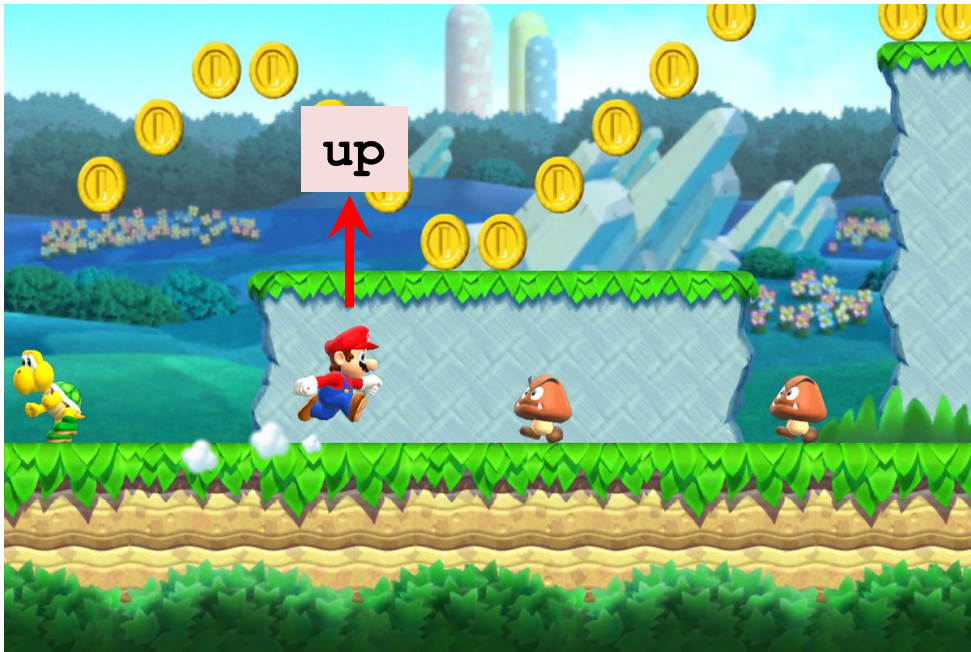
术语: state transition



state transition



术语: state transition

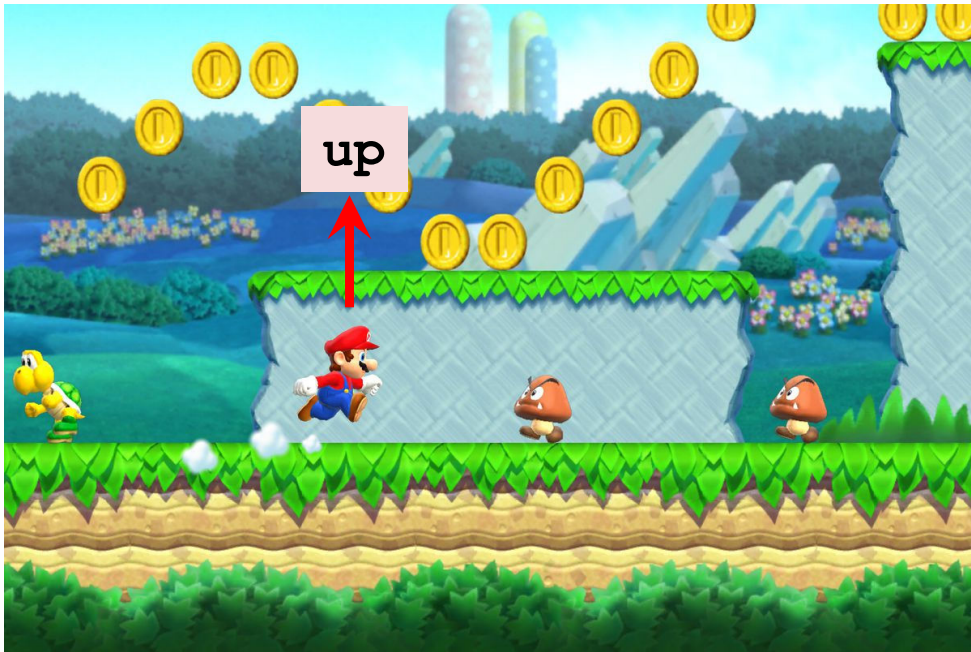


state transition



- E.g., “up” action leads to a new state.

术语: state transition

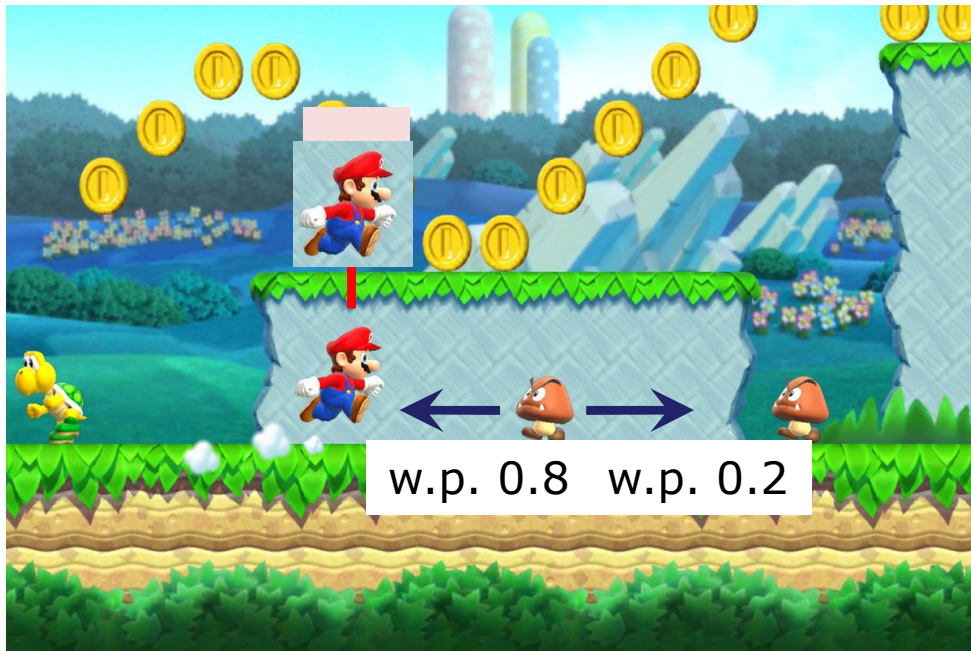


state transition



- E.g., “up” action leads to a new state.
- State transition can be random.
- Randomness is from the environment.

术语: state transition

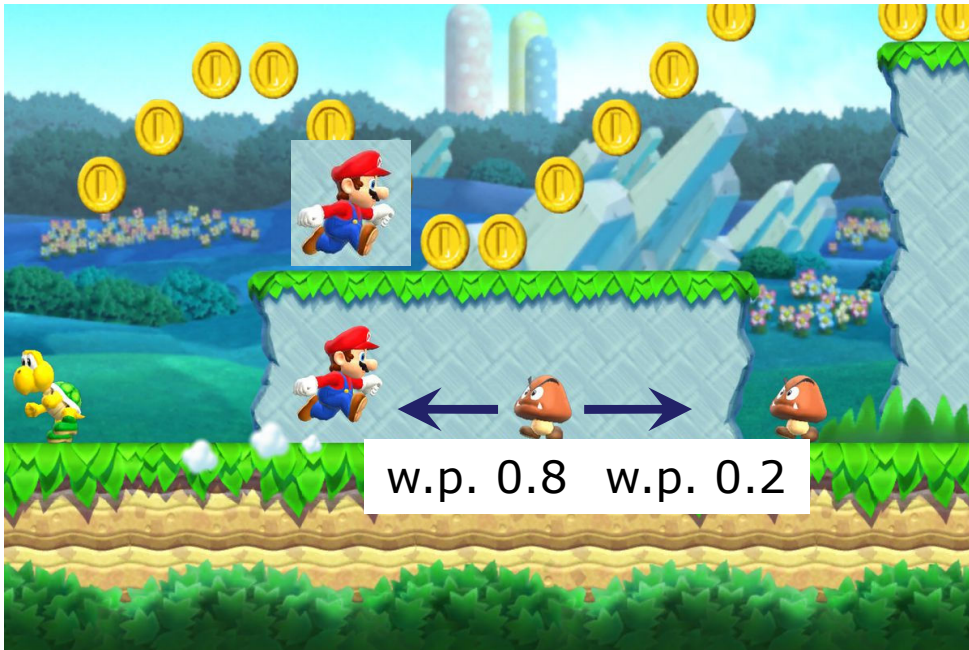


state transition



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术语: state transition



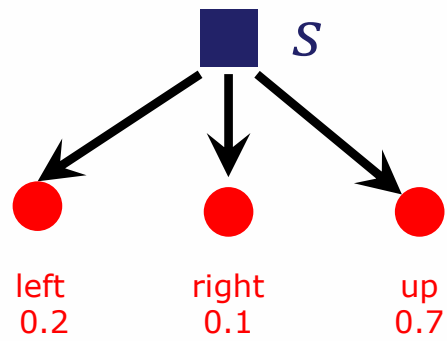
state transition



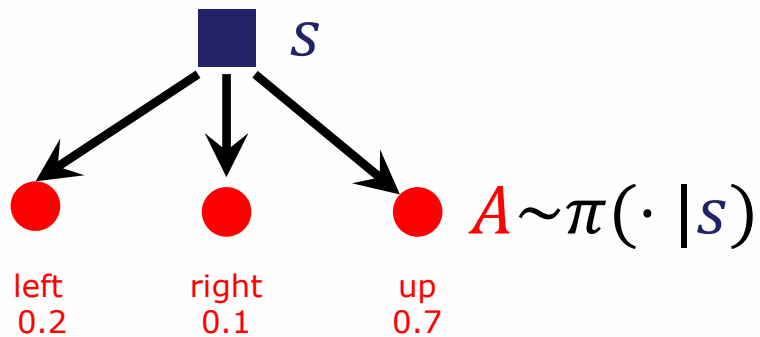
- E.g., “up” action leads to a new state.
- State transition can be random.
- Randomness is from the environment.
- $p(s'|s, a) = \mathbb{P}(S' = s' | S = s, A = a)$.

随机性的两个来源

Randomness in Actions



Randomness in Actions



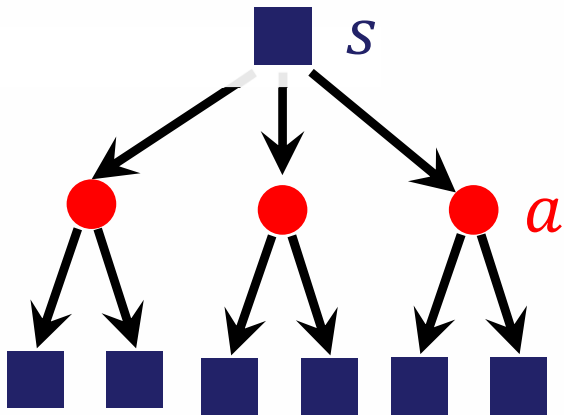
Given state s , the action can be random, e.g., .

- $\pi(\text{"left"}|s) = 0.2,$

- $\pi(\text{"right"}|s) = 0.1,$

- $\pi(\text{"up"}|s) = 0.7.$

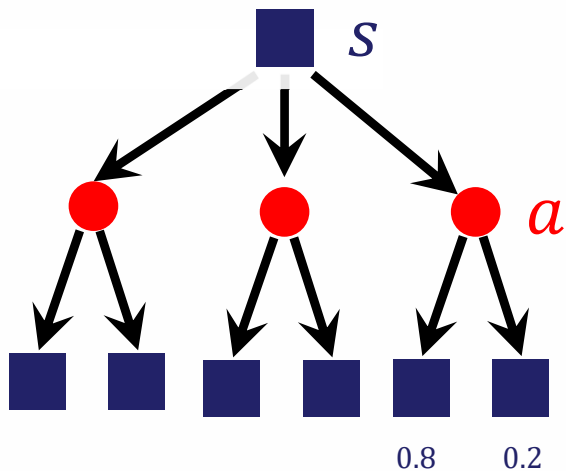
Randomness in States



- State transition can be random.
- The environment generates the new state S' by

$$S' \sim p(\cdot | s, a) .$$

Randomness in States



- State transition can be random.
- The environment generates the new state S' by

$$S' \sim p(\cdot | s, a) .$$

Two Sources of Randomness

- The randomness in **action** is from the policy function:

$$A \sim \pi(\cdot | s) .$$

- The randomness in state is from the state-transition function:

$$S' \sim p(\cdot | s, a) .$$

Rewards and Returns

Return

Definition: Return (i.e., cumulative future reward).

$$\boxed{U_t = R_t + R_{t+1} + R_{t+2} + R_{t+3} + \dots}$$

Return

Definition: Return (aka cumulative future reward).

$$\square U_t = R_t + R_{t+1} + R_{t+2} + R_{t+3} + \dots$$

Question: At time t , are R_t and R_{t+1} equally important?

- Which of the followings do you prefer?
 - I give you \$100 right now.
 - I will give you \$100 one year later.

Return

Definition: Return (aka cumulative future reward).

$$\square U_t = R_t + R_{t+1} + R_{t+2} + R_{t+3} + \dots$$

Question: At time t , are R_t and R_{t+1} equally important?

- Which of the followings do you prefer?
 - I give you \$80 right now.
 - I will give you \$100 one year later.

Return

Definition: Return (aka cumulative future reward).

$$\square U_t = R_t + R_{t+1} + R_{t+2} + R_{t+3} + \dots$$

Question: At time t , are R_t and R_{t+1} equally important?

- Which of the followings do you prefer?
 - I give you \$80 right now.
 - I will give you \$100 one year later.
- Future reward is less valuable than present reward.
- R_{t+1} should be given less weight than R_t .

Discounted Returns

Definition: Return (aka cumulative future reward).

$$\square U_t = R_t + R_{t+1} + R_{t+2} + R_{t+3} + \dots$$

Definition: Discounted return (aka cumulative discounted future reward).

γ : discount factor (tuning hyper-parameter).

- $U_t = R_t + \gamma R_{t+1} + \gamma^2 R_{t+2} + \gamma^3 R_{t+3} + \dots$



Discounted Returns

Definition: Discounted return (at time t).

$$\square U_t = R_t + \gamma R_{t+1} + \gamma^2 R_{t+2} + \cdots + \gamma^{n-t} R_n.$$

Randomness in Returns

Definition: Discounted return (at time t).

$$\square U_t = R_t + \gamma R_{t+1} + \gamma^2 R_{t+2} + \cdots + \gamma^{n-t} R_n.$$

At the end of the game, we observe u_t .



- We observe all the rewards, $r_t, r_{t+1}, r_{t+2}, \dots, r_n$.
- We thereby know the discounted return:

$$u_t = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \cdots + \gamma^{n-t} r_n.$$



Randomness in Returns

Definition: Discounted return (at time t).

$$\square U_t = R_t + \gamma R_{t+1} + \gamma^2 R_{t+2} + \cdots + \gamma^{n-t} R_n.$$

At time t , the rewards, $R_t, \dots, R_n$, are **random**, so the return U_t is **random**.

Randomness in Returns

Definition: Discounted return (at time t).

$$\square U_t = R_t + \gamma R_{t+1} + \gamma^2 R_{t+2} + \cdots + \gamma^{n-t} R_n.$$

At time t , the rewards, $R_t, \dots, R_n$, are **random**, so the return U_t is **random**.

- Reward R_i depends on S_i and A_i .
- States can be random: $S_i \sim p(\cdot | s_{i-1}, a_{i-1})$.
- Actions can be random: $A_i \sim \pi(\cdot | s_i)$.
- If either S_i or A_i is random, then R_i is random.

Randomness in Returns

Definition: Discounted return (at time t).

$$\square U_t = R_t + \gamma R_{t+1} + \gamma^2 R_{t+2} + \dots + \gamma^{n-t} R_n.$$

At time t , the rewards, $R_t, \dots, R_n$, are **random**, so the return U_t is **random**.

- Reward R_i depends on S_i and A_i .
- U_t depends on $R_t, R_{t+1}, \dots, R_n$.
- $\rightarrow U_t$ depends on $S_t, A_t, S_{t+1}, A_{t+1}, \dots, S_n, A_n$.

Value Functions

Action-Value Function $Q_{\pi}(s, a)$

Definition: Discounted return.

$$\square U_t = R_t + \gamma R_{t+1} + \gamma^2 R_{t+2} + \cdots + \gamma^{n-t} R_n.$$

Action-Value Function $Q_{\pi}(s, a)$

Definition: Discounted return.

$$\square U_t = R_t + \gamma R_{t+1} + \gamma^2 R_{t+2} + \dots + \gamma^{n-t} R_n.$$

Definition: Action-value function.

$$\bullet Q_{\pi}(s_t, a_t) = \mathbb{E} [U_t \mid S_t = s_t, A_t = a_t].$$

Action-Value Function $Q_\pi(s, a)$

Definition: Discounted return.

$$\square U_t = R_t + \gamma R_{t+1} + \gamma^2 R_{t+2} + \cdots + \gamma^{n-t} R_n.$$

Definition: Action-value function.

- $Q_\pi(s_t, a_t) = \mathbb{E} [U_t \mid S_t = s_t, A_t = a_t]$.



U_t depends on states $s_t, s_{t+1}, \dots, s_n$ and actions

$a_t, a_{t+1}, \dots, a_n$.



Action-Value Function $Q_\pi(s, a)$

Definition: Discounted return.

$$\square U_t = R_t + \gamma R_{t+1} + \gamma^2 R_{t+2} + \cdots + \gamma^{n-t} R_n.$$

Definition: Action-value function.

$$\bullet Q_\pi(s_t, a_t) = \mathbb{E} [U_t \mid S_t = s_t, A_t = a_t].$$

Regard s_t and a_t as observed values.

Regard $S_{t+1}, \dots, S_n$ and $A_{t+1}, \dots, A_n$ as random variables.



Action-Value Function $Q_\pi(s, a)$

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Definition: Action-value function.

$$\bullet Q_\pi(s_t, a_t) = \square \mathbb{E} [U_t \mid S_t = s_t, A_t = a_t].$$

- $S_{t+1} \sim p(\cdot \mid s_t, a_t),$
 $\vdots$
- $S_n \sim p(\cdot \mid s_{n-1}, a_{n-1}).$

- $A_{t+1} \sim \pi(\cdot \mid s_{t+1}),$
 $\vdots$
- $A_n \sim \pi(\cdot \mid s_n).$

Action-Value Function $Q_{\pi}(s, a)$

Definition: Discounted return.

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Definition: Action-value function.

- $Q_{\pi}(s_t, a_t) = \mathbb{E} [U_t \mid S_t = s_t, A_t = a_t]$.

- $Q_{\pi}(s_t, a_t)$ depends on s_t, a_t, π , and p .
- $Q_{\pi}(s_t, a_t)$ is independent of $S_{t+1}, \dots, S_n$ and $A_{t+1}, \dots, A_n$.



State-Value Function $V_\pi(s)$

Definition: Discounted return.

$$\square U_t = R_t + \gamma R_{t+1} + \gamma^2 R_{t+2} + \dots + \gamma^{n-t} R_n.$$

Definition: Action-value function.

$$\bullet Q_\pi(s_t, a_t) = \mathbb{E} [U_t \mid S_t = s_t, A_t = a_t].$$

Definition: State-value function.

$$\bullet V_\pi(s_t) = \mathbb{E}_A [Q_\pi(s_t, A)]$$

State-Value Function $V_\pi(s)$

Definition: Discounted return.

$$\square U_t = R_t + \gamma R_{t+1} + \gamma^2 R_{t+2} + \dots + \gamma^{n-t} R_n.$$

Definition: Action-value function.

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Definition: State-value function.

$$\bullet V_\pi(s_t) = \mathbb{E}_A [Q_\pi(s_t, A)]$$



Taken w.r.t. the action $A \sim \pi(\cdot | s_t)$.

State-Value Function $V_\pi(s)$

Definition: Discounted return.

$$\square U_t = R_t + \gamma R_{t+1} + \gamma^2 R_{t+2} + \dots + \gamma^{n-t} R_n.$$

Definition: Action-value function.

$$\bullet Q_\pi(s_t, a_t) = \mathbb{E} [U_t \mid S_t = s_t, A_t = a_t].$$

Definition: State-value function.

$$\bullet V_\pi(s_t) = \mathbb{E}_A [Q_\pi(s_t, A)] = \sum_a \pi(a|s_t) \cdot Q_\pi(s_t, a). \quad (\text{Actions are discrete.})$$



Taken w.r.t. the action $A \sim \pi(\cdot | s_t)$.

State-Value Function $V_\pi(s)$

Definition: Discounted return.

$$\square U_t = R_t + \gamma R_{t+1} + \gamma^2 R_{t+2} + \cdots + \gamma^{n-t} R_n.$$

Definition: Action-value function.

$$\bullet Q_\pi(s_t, a_t) = \mathbb{E} [U_t \mid S_t = s_t, A_t = a_t].$$

Definition: State-value function.

- $V_\pi(s_t) = \mathbb{E}_A [Q_\pi(s_t, A)] = \sum_a \pi(a|s_t) \cdot Q_\pi(s_t, a).$ (Actions are discrete.)
- $V_\pi(s_t) = \mathbb{E}_A [Q_\pi(s_t, A)] = \int \pi(a|s_t) \cdot Q_\pi(s_t, a) da .$ (Actions are continuous.)



Understanding the Value Functions

- **Action**-value function: $Q_{\pi}(s, a) = \mathbb{E} [U_t | S_t = s, A_t = a]$.
- Given policy π , $Q_{\pi}(s, a)$ evaluates how good it is for an agent to pick action a while being in state s .

Understanding the Value Functions

□ Action-value function: $Q_{\pi}(s, a) = \mathbb{E} [U_t | S_t = s, A_t = a]$.

□ Given policy π , $Q_{\pi}(s, a)$ evaluates how good it is for an agent to pick action a while being in state s .

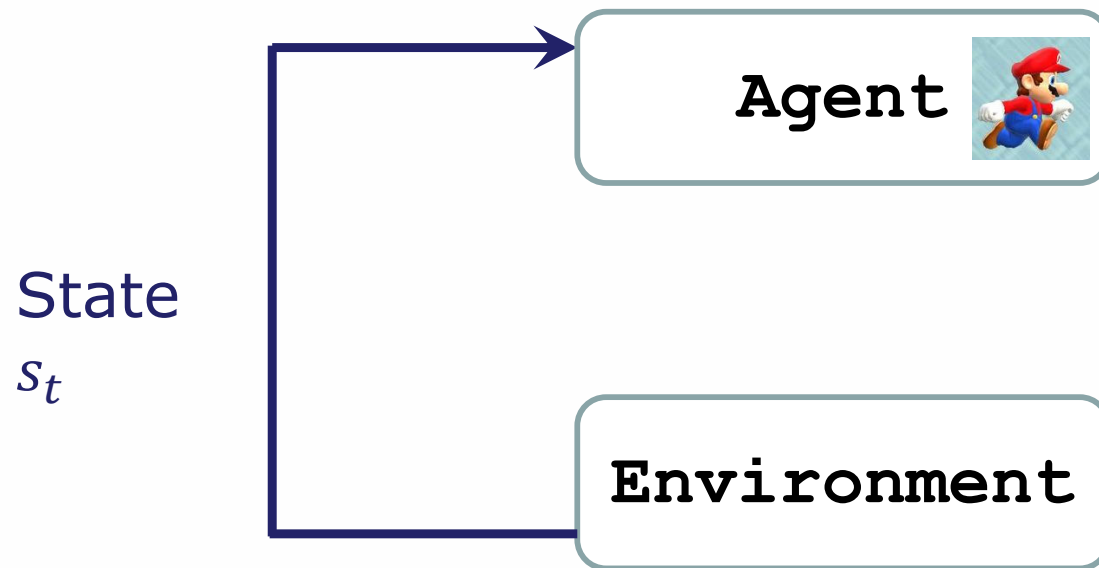
□ State-value function: $V_{\pi}(s) = \mathbb{E}_a [Q_{\pi}(s, a)]$

□ For fixed policy π , $V_{\pi}(s)$ evaluates how good the situation is in state s .

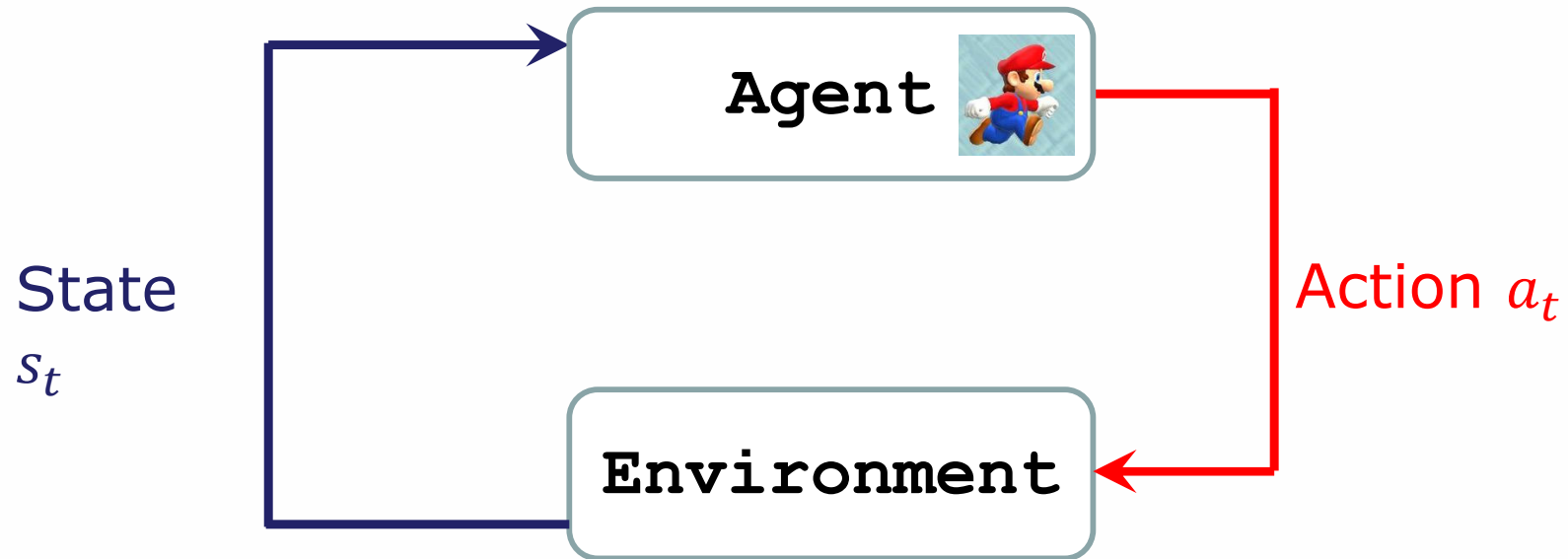
□ $\mathbb{E}_S[V_{\pi}(S)]$ evaluates how good the policy π is.

Agent-Environment Interaction

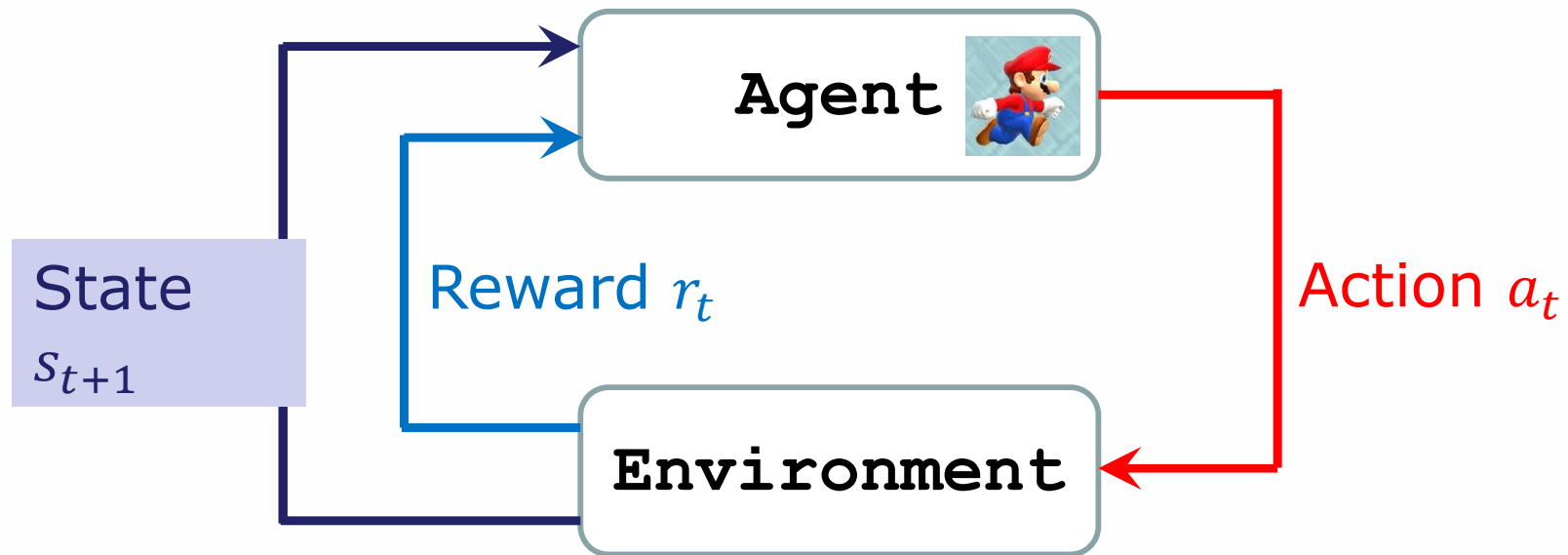
Agent-Environment Interaction



Agent-Environment Interaction



Agent-Environment Interaction

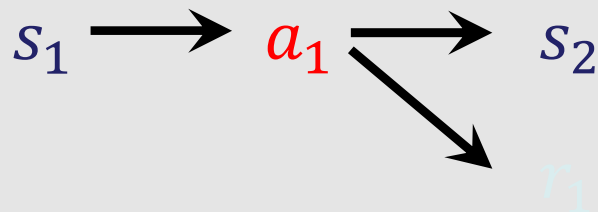


Play game using AI

- Observe state s_t , select action $a_t \sim \pi(\cdot | s_t)$, and execute a_t .
- The environment gives new state s_{t+1} and reward r_t .

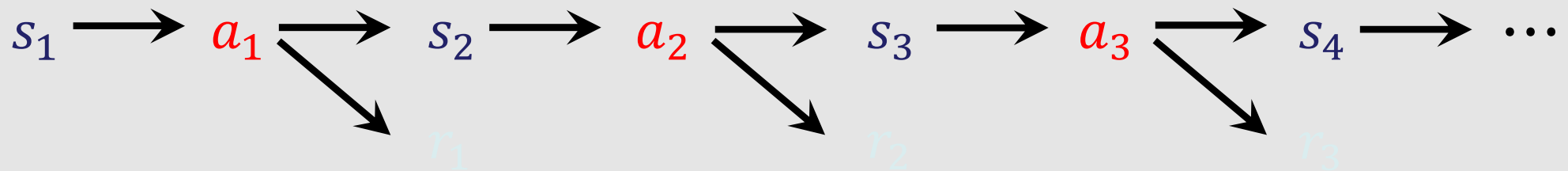
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Play game using AI

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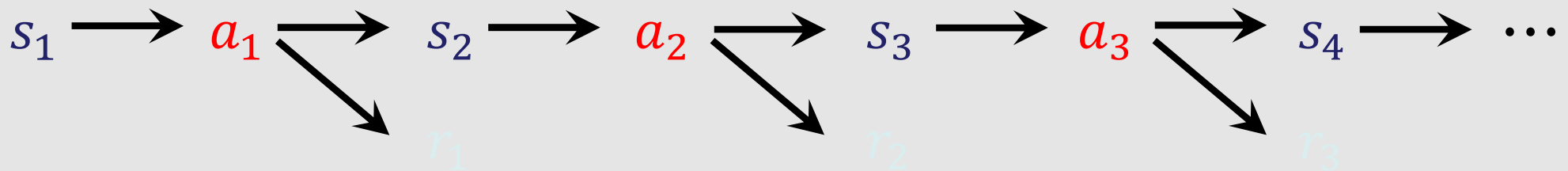


Play game using AI

- (state, action, reward) trajectory:

$s_1, a_1, r_1, s_2, a_2, r_2, \dots, s_n, a_n, r_n.$

- One episode is from the beginning to the end (Mario wins or dies).



Play game using AI

- (state, action, reward) trajectory:

$$s_1, a_1, r_1, \quad s_2, a_2, r_2, \quad \dots, \quad s_n, a_n, r_n.$$

- One episode is from the the beginning to the end (Mario wins or dies).
- What is a good policy?
- A good policy leads to big cumulative reward: $\sum_{t=1}^n \gamma^{t-1} \cdot r_t$.
- Use the rewards to guide the learning of policy.

Evaluating Reinforcement Learning

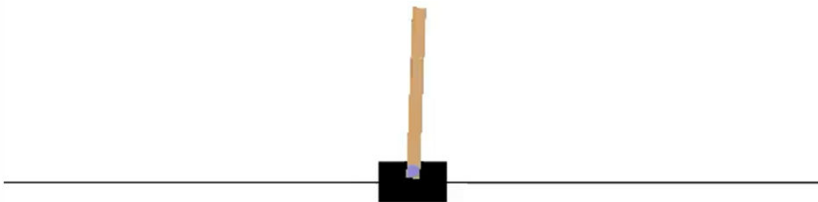
OpenAI Gym

- ❑ Gym is a toolkit for developing and comparing reinforcement learning algorithms.
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Classical control problems



Cart Pole

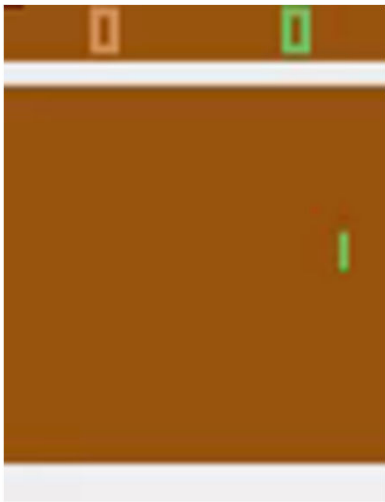


Pendulum

OpenAI Gym

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Atari Games



Pong



Space Invader



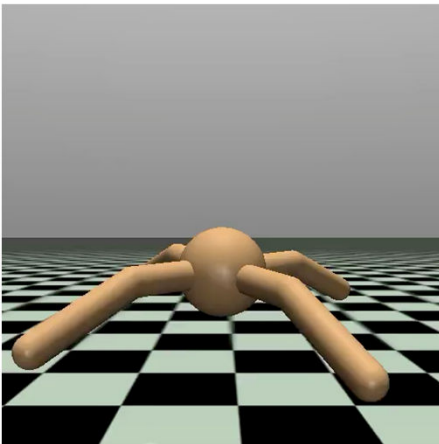
Breakout

OpenAI Gym

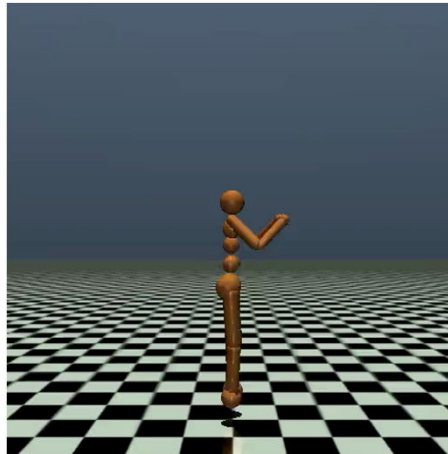
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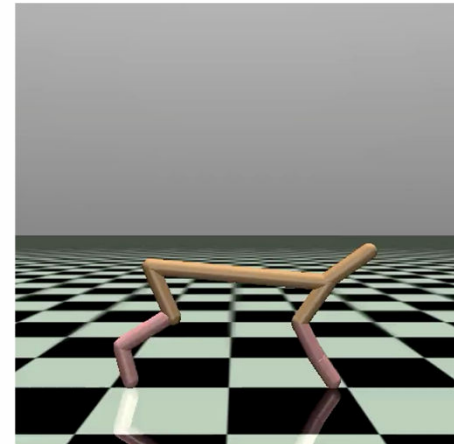
MuJoCo (Continuous control tasks.)



Ant



Humanoid



Half Cheetah

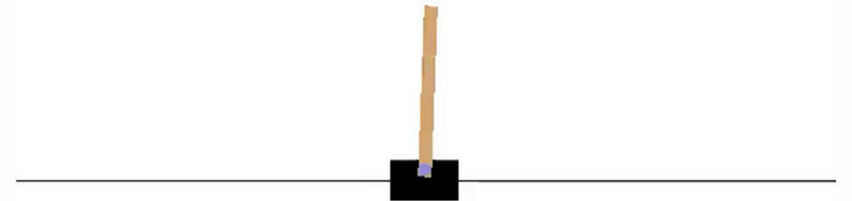
OpenAI Gym



Play CartPole Game

```
➔ import gym  
➔ env = gym.make('CartPole-v0')
```

- ❑ Get the environment of CartPole from Gym.
- ❑ “env” provides states and reward.



Play CartPole Game

```
→ state = env.reset()
→ for t in range(100):
    → env.render()
      print(state)
    → action = env.action_space.sample()
    → state, reward, done, info = env.step(action)
      [ ] [ ]
    → if done:
        print
        break
env.close()
```

A window pops up rendering CartPole.

A random action.

"done=1" means finished (win or lose the game)

Summary

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Terminologies

- Agent 
- Environment
- State s
- Action a
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$$U_t = R_t + \gamma R_{t+1} + \gamma^2 R_{t+2} + \dots$$

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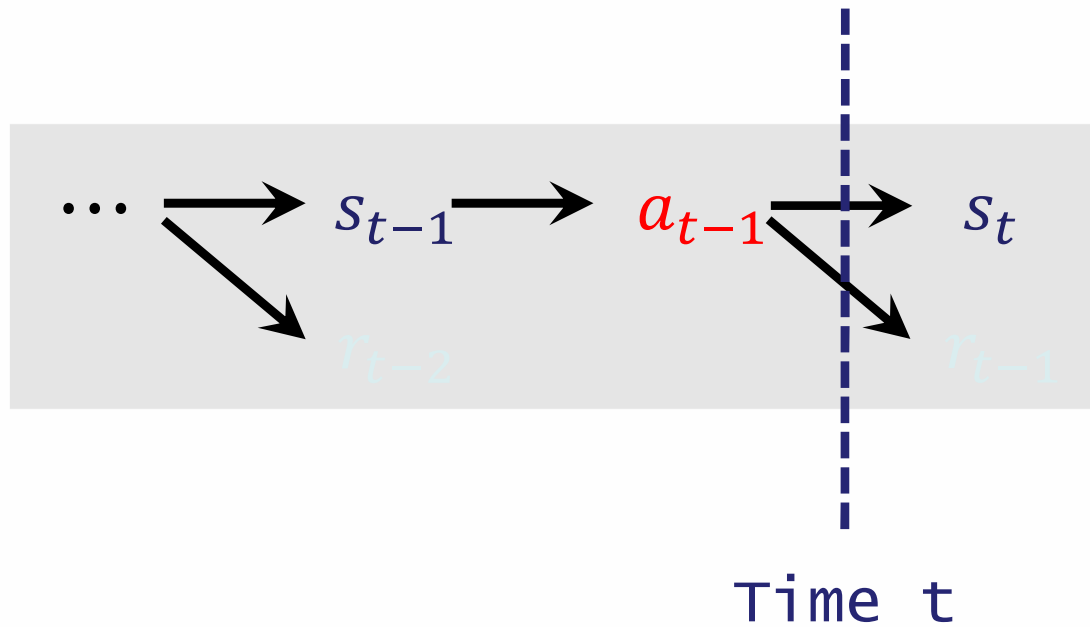
- Action-value function:

$$Q_\pi(s_t, a_t) = \mathbb{E}[U_t | s_t, a_t].$$

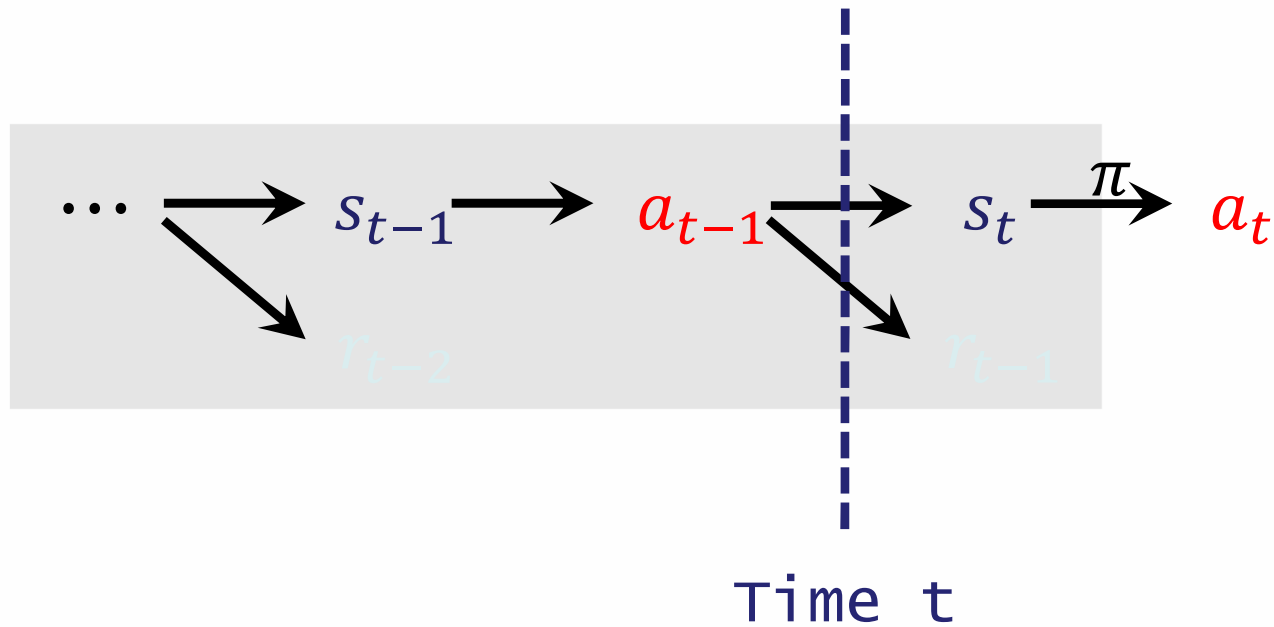
- State-value function:

$$V_\pi(s_t) = \mathbb{E}_{A \sim \pi}[Q_\pi(s_t, A)].$$

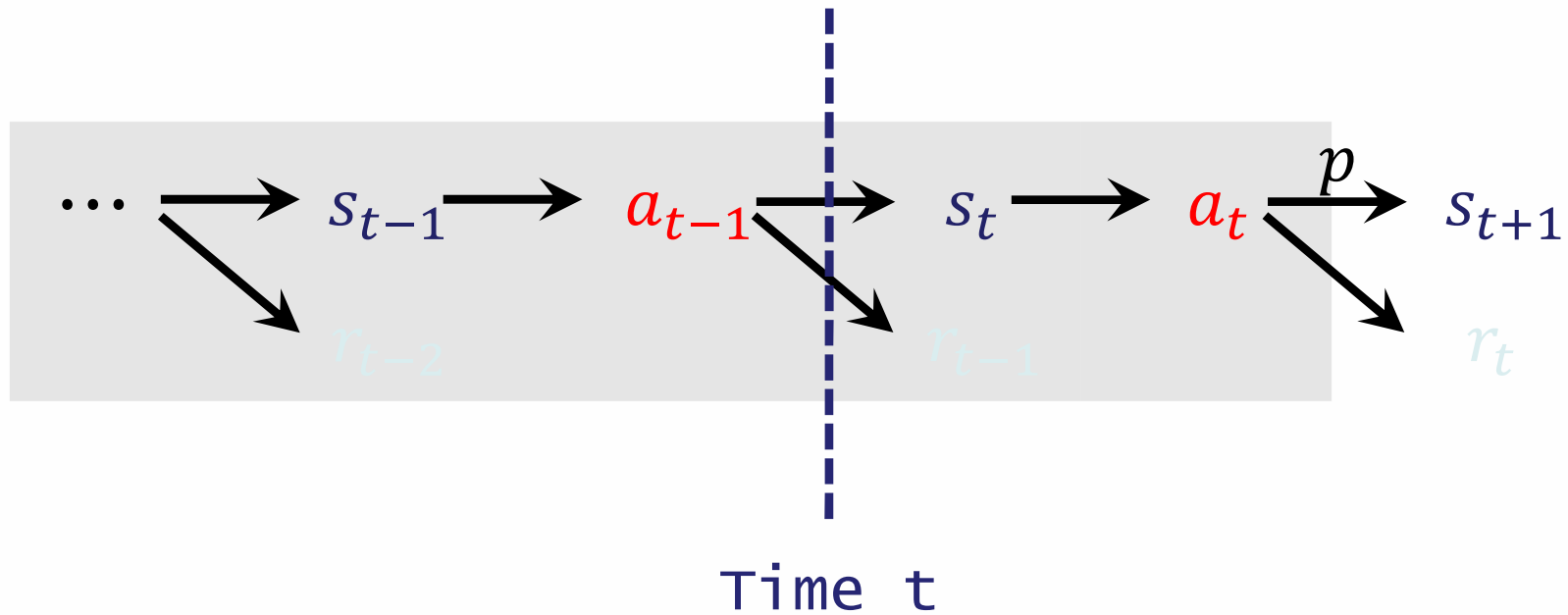
Play game using AI



Play game using AI



Play game using AI



Play game using AI

